Counterfactual Attribution in Cascaded Retrieval

Progress Report, Weeks 1–2 • Anupam Dwivedi, Shubham Paliwal, Chaitanya K
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Which stage of a two-stage cascade is responsible for its output quality? For a *pointwise* reranker, once the candidate set C_q is fixed the final ranking is invariant to the Stage-1 scores: Stage 1's only causal channel is *membership* in C_q — except under a budget, where Stage-1 rank decides who survives the cut. Score-level attribution therefore cannot answer the question, so we build four counterfactual worlds over a frozen universe U_q and read three non-additive estimands off them. Their sign rules hold only *conditionally on membership*, so all validation is stratified and nothing is pooled.

Setup. 97 judged TREC DL 2019+2020 queries on MS MARCO v1 passages. Stage 1 is BM25 or cosDPR-distil (exact dense search) to depth 5,000, Stage 2 the MiniLM-L-12 cross-encoder; U_q is the union of both runs, C_q a system's top K_c. E_{rerank}, E_{gate}: a document's actual utility minus its no-rerank or no-gate utility. Δ_t: the change when a fixed budget keeps a share t for C_q and spends the rest on outsiders picked by Stage-1 rank (`boundary`), the other retriever (`alternative`) or at random. H7: do gate-sensitive queries gain more from a lower t than reranking-sensitive ones?

Weeks 1 and 2 are complete, with all three workstreams merged. No real Stage-2 scores exist yet; that is Week 3.

STATE

Config fingerprint	`46d0807d8ed3` (v1 = v2)
Queries / judgments	97 / 20,646
$\lvert U_q \rvert$	7,045–9,986 (med. 9,097)
Universe pairs	**872,455**
Headline K_c	100
Allocations audited	11,640
Stage-2 bill	872,455 pairs, 52–57 min
Tests passing	**442**
Commits / merged PRs	46 / 4

Universe reproduced bit-identical in three environments on two operating systems: most of the Week-4 reproducibility pack, early.

STAGE-1 RUNS VS. PUBLISHED FIGURES

		nDCG@10		R@1000	
		ours	pub.	ours	pub.
BM25	DL19	0.5058	0.5058	0.7501	0.7501
	DL20	0.4796	0.4796	0.7863	0.7863
DPR	DL19	0.7063	0.7250	0.8224	0.8201
	DL20	0.6889	0.7025	0.8587	0.8533

BM25 matches exactly: the run is *right*, not just well formed. DPR is 0.014–0.019 under Pyserini's exact-search nDCG@10, above its recall; float32 re-scoring gave the same nDCG, so float16 is not the cause.

1. Work completed

Shared (224 tests): world and allocation builders, effects, membership strata, stratified metrics, schemas, provenance. **Anupam** (145 tests): loaders and the by-query split, retriever interface, BM25 run and verifier, the frozen universe, the feasibility gate, Stage-1 over U_q, the reference subset and the t-allocations with audits. **Shubham** (46 tests): DPR Stage-1 run, the counterfactual engine, E1 exact-recovery validation, the H7 power simulation, effect analysis, condensed-mode support. **Chaitanya** (27 tests): reranker wrapper, cache filler, features, predictors, cost–fidelity harness, shard plan, GPU throughput and the shuffle-invariance test.

2. Key results

Feasibility gate passed. Candidate Jaccard 0.11–0.16, so the two Stage-1 systems select genuinely different documents and the cross-paradigm comparison has leverage. The expected confound — TREC pools favouring lexical runs — does *not* exist: judged density of C_q is 1.06× between systems, and the Stage-1 floor is symmetric at an identical 0.450 for both. Outsider-policy supply is 40–49× the budget, so `boundary` and `alternative` are real policies rather than tie-breaks in disguise.

Cost is not a constraint. The cache keys on (query, doc) and every world re-ranks a subset of U_q, so the whole project costs 872,455 calls *once* — 52–57 minutes on one T4 at 257–282 pairs/sec, measured on two real-passage shard fills. This retired the reference-subset sampling design. Shuffle-invariance passed on the real reranker, verifying rather than assuming the pointwise property the E_{gate} derivation rests on.

H7 is confirmatory for one policy only. At K_c=100 and the real $\lvert U_q \rvert$, only all 97 queries reach 0.80 power, and then only for `alternative` (0.935/0.875); `boundary` falls short (0.67/0.41). Declared descriptive in advance.

3. Corrections — the fortnight's most valuable output

Six defects were found and fixed, **four of them in checks that passed while being structurally unable to detect the failure they existed to catch**: the H7 power denominator conditioned on testability and so overstated power; "relevant" counted grade-1 documents that carry zero gain, inflating the gate-failure supply 4–9× and nearly deciding the headline K_c from the wrong number; the BM25 verifier used exponential gain where `trec_eval` uses linear, so it would have "verified" against an unreachable number; and the allocation spend check compared against the amount *achieved*, so a short allocation matched its own shortfall. Separately, our own pre-declared headline-K_c rule conflated a non-degeneracy floor with a selection rule and so chose the depth with *least* signal; the correction is recorded as post-hoc, defensible only because supply is measurable before any effect exists. The sixth was feature leakage: Stage-2 features over all of U_q do not *approximate* e_{gate} for a non-survivor, they *determine* it, since a cached score outside C_q exists only because world C was computed. Now restricted to survivors, enforced by a renamed parameter.

4. Open validity threat, and how it is handled

U_q is 1.65% judged while BM25's top-100 is 51% judged. World C sorts ∼9,000 mostly-unjudged documents, so judged survivors are displaced out of its top-10 and E_{gate} drifts towards DCG(π_{act}) as an artefact of 2019 pooling. Resolved *procedurally* at freeze v2: every effect is computed both uncondensed and condensed (Sakai, SIGIR 2007). Condensing transforms rankings, not scores, so it costs no extra GPU calls, and it is the only option that does not decide before the evidence exists. The uncondensed estimand stays primary; a disagreement is itself a finding.

5. Future work

Week 3. Nothing is blocked. Regenerate the synthetic fidelity numbers under the corrected feature boundary; fill the cache; compute E_{rerank}, E_{gate} and Δ_t over the real universe in both modes; and take the decisive measurement — if world C's top-10 is ∼20% unjudged the concern is a footnote, if ∼95% the condensed figure becomes the headline. **Week 4.** Effect distributions by stratum, the cost–fidelity curve (converging to the exact reference at full budget), the H7 comparison, predictor fidelity against all three baselines, and the reproducibility pack.

Known risks. Judgment sparsity is mitigated procedurally, not solved; H7 is descriptive for `boundary` at any feasible size; and gate-failure supply is thin even at the headline depth (median 6, with 17–23 of 97 queries having none), so per-query distributions will be noisy and are reported with their supply.

Repository `a-n-u-p-a-m/IRE_PROJECT`; decisions and an extended report in `docs/`.